Sikāi Sathi — UX/UI Audit Report

Nov 2, 2026 · @nirajan

Executive summary

Overall UX score: 5.5 / 10. Sikāi Sathi has the right idea — diagnose the exact step, explain in Nepali, prove learning with practice — and a clean, fast, accessible-by-default base. But the demo build undermines its own promise at three points: the diagnosis is canned, the Nepali experience stops at one paragraph, and the result screen says "Mastery: YES" under a wrong answer.

Dimension	Score /10	One-line verdict
Concept & value proposition	8	Step-level diagnosis + mother-tongue explanation is a sharp, fundable idea
Learning design	4	No retry, answer revealed, mastery logic contradicts the result
Diagnosis trust	3	Every wrong input gets "forgot to divide" in the mock build
Information architecture	6	Clear linear wizard, but 7 screens where 4 would do
Visual design	6	Tidy and consistent; red overused; Nepali visually demoted
Content & localisation	4	Nepali selected, yet diagnosis, result and buttons stay English
Accessibility (WCAG 2.1 AA)	5	6 criteria fail; all cheap to fix
Mobile readiness	5	Reflows, but CTA below the fold and photo capture is weak
Performance	9	84 ms first paint, 15.7 KB page; needs waiting states for real AI latency

Top 5 fixes, in order

1. Make the misconception detection real and deterministic for the 3 supported errors, with an "I couldn't read that" state.
2. Replace reveal-and-end with hint → retry → worked solution, and redefine mastery so it never contradicts the result.
3. Localise every UI string, and give Nepali a proper Devanagari font and first-class size.
4. Cut 7 screens to 4, pin the equation on every screen, and highlight the wrong step.
5. Fix the 6 WCAG failures and the phone layout (sticky CTA, camera capture, photo resize).

Items 1–3 alone move the score to about 7.5; all five to about 8.5.

Scope & method

The audit covers the full 7-screen flow of the local build at `127.0.0.1:8000`, walked end to end with live API calls, plus a source review of its single HTML page (15.7 KB, one inline stylesheet, one inline script).

Item	Detail
Screens	1 Upload · 2 Steps · 3 Mistake · 4 Explain · 5 Visual · 6 Practice · 7 Result
Heuristics	Nielsen's 10 usability heuristics; learning-design principles (feedback, retrieval practice, scaffolding)
Accessibility	WCAG 2.1 AA: contrast computed from the CSS tokens, accessibility tree, keyboard and screen-reader semantics
Viewports	1435 px desktop, 768 px tablet, 375 px phone
Paths tested	Default demo ("forgot to divide"), correct steps, wrong practice answer, empty input, English vs Nepali

Severity scale

- **S1 Critical** — blocks learning or misleads the student; fix before any demo

- **S2 Major** — causes confusion, drop-off or an accessibility failure
- **S3 Minor** — friction or polish
- **S4 Enhancement** — opportunity, not a defect

Product & users

Sikāi Sathi diagnoses a student's handwritten or typed algebra steps, names the misconception, explains it in Nepali or English, shows a balance-scale visual, then checks one practice answer. Correctness is computed by deterministic Python; the LLM only explains.

Primary user (assumed): a Nepali secondary student, Grade 7–10, solving one-variable linear equations, most likely on a low-to-mid Android phone with patchy data. Comfortable reading Nepali; English is a second language.

Secondary users (assumed): a teacher or parent who wants to see what the student got wrong and whether they improved; a hackathon judge seeing the demo for 2–3 minutes.

Job to be done	How the app serves it today	Gap
"Tell me *where* I went wrong"	Lists detected steps and an evidence line	Wrong step is not highlighted; evidence is written in algebra notation, not student language
"Help me understand *why*"	Nepali explanation + static balance visual	Explanation ends in a question with no place to answer it; visual is not tied to the student's own error
"Let me prove I've got it"	One practice question, final answer only	No retry, answer revealed on first miss, steps not checked
"Show me I'm improving"	Raw counters and a Mastery flag	Mastery flag contradicts the result shown on the same screen

Information architecture & navigation

The app is a strict linear 7-screen wizard; it is easy to follow but three screens carry two lines of content each, and the student loses sight of the problem after screen 1.

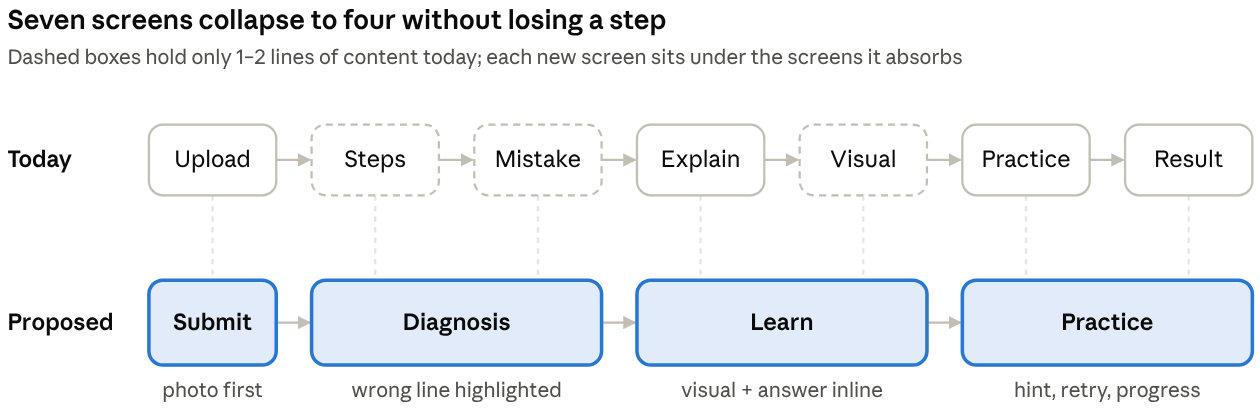
What works

- One card at a time keeps cognitive load low for a young learner.
- Every screen has a Back button, and the stepper marks done (green), current (red) and upcoming (grey).
- Primary action always sits bottom-left in the same place.

Problems

1. **Too many steps for one learning loop (S2).** Screen 2 (Detected steps) and screen 3 (Mistake) together show 3 equation lines and 2 sentences. Screen 5 (Visual) is a single static picture. Each extra click is a drop-off point on a phone.
2. **The problem disappears (S2).** From screen 2 onward, nothing restates which equation the student is working on. On screen 6 a different equation appears (4x + 6 = 30) with no bridge sentence.
3. **The correct path is routed through "mistake" screens (S2).** A student who solves it correctly still taps "What did I get wrong?" and lands on "3 · The misconception" to read "No mistake found".
4. **Stepper is display-only (S3).** Chips are not links, so a student cannot jump back to the explanation from Practice without pressing Back twice. Chips have no `aria-current`.
5. **Start over reloads the page (S3).** It wipes Student ID, language and problem choices, so every retry starts from defaults.
6. **No entry or exit context (S3).** There is no one-line "what this does" before the form, and no history view, although `/api/progress` already returns a `recent` activity list.

Recommended structure: collapse to 4 screens — **Submit → Diagnosis** (steps with the wrong line highlighted + misconception name) **→ Learn** (explanation, visual and the follow-up question answered inline) **→ Practice** (answer, retry, result and progress). Pin the original equation in a slim bar under the stepper on every screen.



current vs proposed flow · 7 screens to 4

The three dashed screens are the ones that merge away; a student reaches practice in 3 taps instead of 5.

Visual design

The UI is clean, token-based and consistent, but red does three jobs at once (action, current step, wrong answer) and Nepali text is styled as secondary in a product that promises "your language".

Element	Current	Verdict	Recommendation
Colour system	9 CSS tokens; crimson `#7f1d1d` header, `#b91c1c` buttons, slate neutrals	Good base; crimson is culturally apt	Keep crimson for brand only. Move primary buttons to Nepal-flag blue `#003893` or a deep teal; reserve amber for "not yet" and green for "correct"
Red overload	Primary CTA, active step chip, error banner, both balance pans (`#fee2e2`) are red	**S2** — a mistake-diagnosis tool for children that looks like a teacher's red pen raises anxiety	Errors in amber with a growth-mindset tone; balance pans neutral, wrong side amber
Latin typeface	Noto Sans, 16 px body, 1.55 line height	Good	Keep
Devanagari typeface	Not declared; browser falls back to whatever system font is installed	**S2** — weight and x-height jump mid-line in "Upload your work फोटो अपलोड गर्नुहोस्"	Load Noto Sans Devanagari (or Mukta) with `unicode-range`; size it 1.05–1.1× the Latin size
Nepali hierarchy	Nepali sub-labels use `.hint` : 13.6 px, grey	**S2** — the student's first language is visually demoted	When language = Nepali, Nepali is the primary line at full size; English becomes the small secondary line

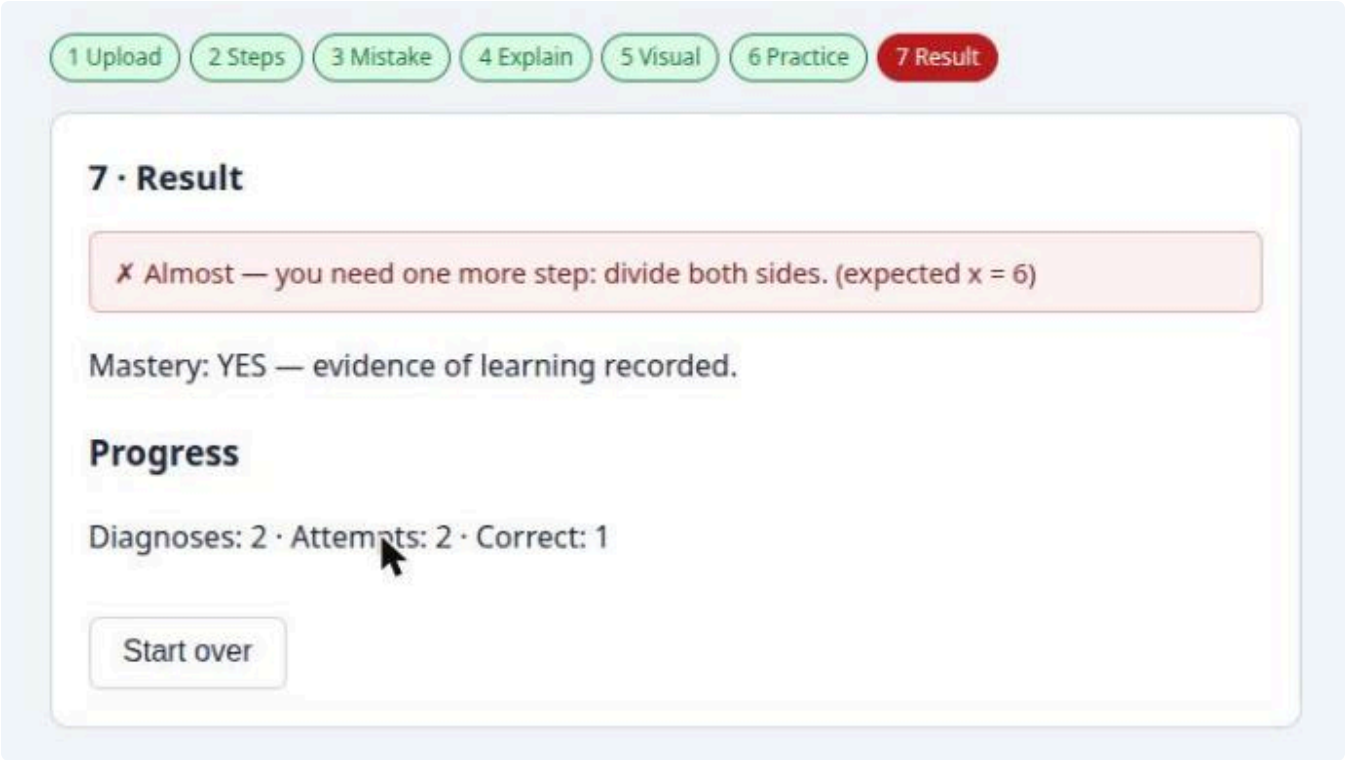
Element	Current	Verdict	Recommendation
Type scale	h1 21.6 px, h2 18.4 px, body 16 px, chips 12 px	Weak contrast between h2 and body	Use a 1.25 scale: h2 22–24 px, body 16–17 px, chips 13 px minimum
Math rendering	Monospace with ASCII hyphen; "Solve: 4x + 6 = 30" set as code	Reads as programming, not maths	Proportional math font, true minus (−), italic *x*; "Solve" as a label, equation large
Steps list	Numbers hang 13 px outside the grey box	**S3** misalignment	`list-style-position: inside` or 2 rem left padding
Native controls	Default grey selects, "Choose File / No file chosen" button 21 px tall	**S3** — looks unfinished, English-only	Custom select style; a large "Take a photo" drop zone
Debug chrome	Dashed amber "provider: mock" tag on screens 2 and 4	**S2** in front of students or judges	Hide behind a `?debug` flag
Balance-scale visual	Static, level, no fulcrum; pole crosses the beam like a T	**S2** — metaphor is right, drawing doesn't teach	Show the student's answer tipping the scale, then animate both pans as −5 and ÷3 are applied
Brand personality	No mascot, icon or illustration; "Sathi" (companion) is only a word	Missed opportunity	A small friendly companion character and warm micro-illustrations on success

Interaction & key flows

The happy-path demo works in 6 taps, but the learning loop breaks at the end: a wrong practice answer reveals the solution, offers no retry, and is followed by "Mastery: YES".

Journey map — default demo, wrong practice answer

Screen	Student does	Student sees	Friction	Likely feeling
1 Upload	Taps Analyze (demo steps pre-filled)	5 fields; CTA below the fold on a 375 px phone	Student ID asked first; photo and typed steps both live, unclear which is used	Unsure
2 Steps	Reads 3 lines	Steps list, "provider: mock", algebraic evidence "(c − b) without dividing by a"	Wrong line not highlighted	Confused
3 Mistake	Reads 1 sentence	"Forgot to divide" + "Confidence: 72%"	Thin screen; confidence number means nothing to a child	Judged
4 Explain	Reads Nepali paragraph	Explanation ending in "2x + 7 = 15 मा x कति हुन्छ?"	Question has no answer box	Stuck
5 Visual	Looks at scale	Static, level balance labelled 3x + 5 | 20	Doesn't show *their* mistake	Neutral
6 Practice	Types 24, presses Enter	Nothing happens; must tap the button	Different equation from the one just asked (4x + 6 = 30)	Mildly lost
7 Result	Reads	"✗ Almost… (expected x = 6)" then "Mastery: YES"	Answer revealed, no retry, contradictory mastery	Mixed signals



Screen 7 after answering 24 to 4x + 6 = 30: the ✗ banner and "Mastery: YES" sit one line apart.

Edge cases tested

Input	Expected	Actual	Severity
Steps `hello`	"I couldn't read that"	Diagnosed *forgot to divide*, 72%, evidence quotes `x = 15` the student never wrote	S1
4x − 3 = 17 → 4x = 14 (sign error)	*Sign error*	*Forgot to divide*; explanation says "after 3x = 15, divide by 3"	S1
3x + 5 = 20 → 3x = 20 (one-sided)	*One-sided operation*	*Forgot to divide*	S1
Problem 2x + 7 = 15 selected, default 3x + 5 steps left in box	"Your steps don't match this problem"	Diagnosed normally; mismatch ignored	S2
Correct steps (x = 5)	Praise, skip to practice	Routed via "What did I get wrong?"; tutor still asks "how would you remove 5?"	S2

Input	Expected	Actual	Severity
Empty steps, no photo	Inline "Add a photo or type your steps"	Banner "Diagnosis failed: 422 — Provide image_base64 or steps_text"	S2
Empty practice answer	Inline "Type a number first"	Banner "Check failed: 422 — [object Object]"	S2
Wrong practice answer	Hint + try again	Answer revealed, flow ends	S1



Screen 6, empty answer submitted: the raw validation error reaches the student.

The three S1 diagnosis rows come from the mock provider, which returns one canned result for every wrong input. The demo currently runs on it, so this is what a judge or student sees. Note the footer's claim — "correctness checked by deterministic Python" — holds for right/wrong answers but not for the misconception label.

Other interaction gaps

- **Loading is invisible (S2).** The "Analyzing…" text sits inside the disabled red button at 1.17:1 contrast. With real OCR + LLM latency of several seconds, students will tap again or leave.
- **Enter does nothing (S2).** No `<form>` wraps the inputs, so the phone keyboard's Go key is dead on screens 1 and 6.
- **Focus is lost on every screen change (S2).** After each transition focus lands on `<body>`; keyboard and screen-reader users must re-find their place.

- **Pre-filled demo data (S2).** The textarea ships with the demo steps. A student who adds a photo still sends those steps.
- **Shared default identity (S3).** Every user who keeps "demo-student" writes to the same progress record.

Content & UX copy

With Nepali selected, only the tutor paragraph and two hint lines change language; the diagnosis, the result, every button and every error stay in English. That breaks the tagline's "Your language" promise at the two moments that matter most.

What works: first-person action labels ("Explain it to me", "Show me visually", "Your turn") are warm and clear. Nepali is the default language. The tutor uses the textbook term पक्ष (side).

Problems

- **Core feedback not localised (S1).** Screen 3's misconception, screen 7's result and mastery lines are English-only.
- **System voice leaks to students (S2).** "provider: mock", "Confidence: 72%", "Evidence: … matches (c − b) without dividing by a", "evidence of learning recorded", "422 — [object Object]".
- **Deficit framing (S2).** "What did I get wrong?" and "The misconception" assume failure, even on a correct submission.
- **Register (S3, test it).** The tutor uses formal तपाईं. An app named *Sathi* (friend) for school students may land warmer with तिमी. Validate with 5–6 students and a teacher.
- **Numerals (S4).** Nepali-medium textbooks often use Devanagari numerals; consider a toggle.
- **Demo leftovers (S3).** "Demo tip" and the developer footer are visible to everyone.

Microcopy rewrites (Nepali in तिमी register)

Where	Current	Proposed (English)	Proposed (Nepali)
Screen 1 field	Student ID	Your name or class code	तिम्रो नाम वा कक्षा कोड
Screen 1 upload	Photo of handwritten work / Choose File	Take a photo of your copy	कपीको फोटो खिच

Where	Current	Proposed (English)	Proposed (Nepali)
Screen 1 CTA	Analyze my work	Check my steps	मेरो चरण जाँच
Screen 2 CTA	What did I get wrong?	See what happened	के भयो हेरौं
Diagnosis	Forgot to divide — subtraction done, final division missed.	You subtracted correctly! One step left: divide both sides by 3.	घटाउने काम ठिक भयो! अब एउटा चरण बाँकी: दुवै पक्षलाई 3 ले भाग गर।
Evidence	Final step 'x = 15' matches (c − b) without dividing by a.	Look at line 3: 3x became x, but 15 wasn't divided by 3.	लाइन 3 हेर: 3x बाट x भयो, तर 15 लाई 3 ले भाग गरिएन।
Confidence	Confidence: 72%	*(teacher view only)*	—
Wrong answer	✗ Almost — you need one more step: divide both sides. (expected x = 6)	Almost! Divide both sides once more and try again.	झन्डै! दुवै पक्षलाई भाग गरेर फेरि हेर।
Mastery	Mastery: YES — evidence of learning recorded.	1 of 3 solved — 2 more to master this skill	3 मध्ये 1 सही — यो सीप पक्का गर्न अझै 2
Empty steps	Diagnosis failed: 422 — Provide image_base64 or steps_text	Add a photo of your work or type your steps.	आफ्नो कामको फोटो खिच वा चरणहरू टाइप गर।
Empty answer	Check failed: 422 — [object Object]	Type your answer first.	पहिले उत्तर लेख।
Loading	Analyzing… (invisible)	Reading your handwriting…	तिम्रो लेखाइ पढ्दैछु…

Accessibility audit

The app does not meet WCAG 2.1 AA today: 6 criteria fail, 2 are partial. The foundations are good (native controls, real `<label for>`, `<ol>`, headings, reduced-motion rule), so most fixes are a few lines each.

WCAG 2.1 criterion	Result	Evidence	Fix
1.4.11 Non-text contrast	**Fail**	Input, select, textarea and Back-button borders `#cbd5e1` on white = 1.48:1 (needs 3:1)	Border `#64748b` (4.8:1)
3.1.2 Language of parts	**Fail**	`<html lang="en">`; no Devanagari string carries `lang="ne"`	Wrap Nepali text in `lang="ne"`; set page `lang` from the language selector
4.1.3 Status messages	**Fail**	Error, info and result banners have no `role` or `aria-live`	`role="alert"` on errors, `role="status"` on result and loading
2.4.3 Focus order	**Fail**	After every screen change `document.activeElement` is `<body>`	Move focus to the new card's `h2` (`tabindex="-1"`)
3.3.1 Error identification	**Fail**	Errors appear in a page-top banner, not tied to the field, as HTTP codes	Inline message under the field + `aria-describedby` + `aria-invalid`
1.4.1 Use of colour	**Fail**	Stepper states differ only by colour (green / red / grey)	Add ✓ to done chips, bold + `aria-current="step"` on current
2.4.7 Focus visible	Partial	Only the browser's thin default ring; low contrast on the red button	`:focus-visible { outline: 3px solid #1d4ed8; outline-offset: 2px }`
1.1.1 Non-text content	Partial	Balance SVG label is generic and static, whatever the problem	Generate the label from the equation and step
1.4.3 Text contrast	Pass*	Body, labels, buttons and chips all ≥ 4.5:1 (table below)	*Loading text inside the busy button is 1.17:1; move it outside
1.3.1, 2.1.1, 1.4.10, 3.3.2	Pass	Labels linked; all controls keyboard-reachable; no horizontal scroll at 375 px	—

Measured contrast (WCAG relative luminance, computed from the CSS tokens)

Pair	Ratio	AA
Body text `#1e293b` on white	14.63:1	Pass
White on header `#7f1d1d`	10.02:1	Pass
Muted text `#475569` on page `#f1f5f9`	6.92:1	Pass
White on button `#b91c1c`	6.47:1	Pass
Done chip `#15803d` on `#dcfce7` (12 px)	4.57:1	Pass, barely
Field border `#cbd5e1` on white	1.48:1	Fail (non-text)
"Analyzing…" `#475569` on red, busy	1.17:1	Fail

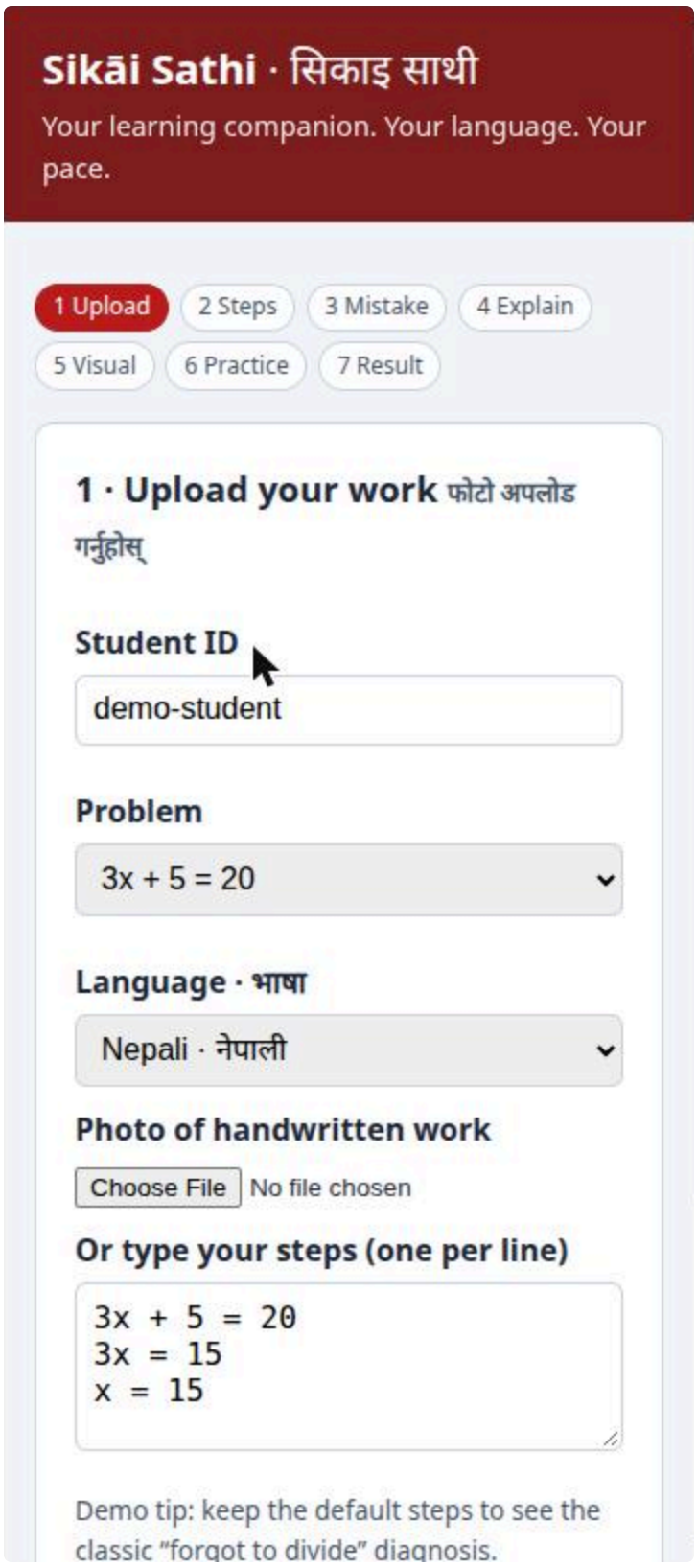
WCAG 2.2 note: the native file-picker button is 21 px tall, under the 24 px minimum of 2.5.8 Target Size.

Responsive & mobile

The layout reflows without horizontal scroll at 375 px, but on a phone the main button sits below the fold and the photo path — the one phone users will use — is the weakest part of the screen.

Measure at 375 × 812 px	Value	Verdict
Horizontal overflow	None (scroll width = viewport)	Pass
Header height	113 px (tagline wraps to 2 lines)	Too tall; 14% of the screen
Stepper	2 rows, 56 px	Use a compact "Step 1 of 4" bar + progress line
"Analyze my work" top edge	844 px	Below the fold — student must scroll to start
Buttons / inputs height	37 px / 38 px	Below the 44–48 px touch norm

Measure at 375 × 812 px	Value	Verdict
File picker height	21 px, native "Choose File"	Hard to tap; no direct camera



Screen 1 at 375 px: the first viewport ends before the "Analyze my work" button.

Phone-specific issues

- **No direct camera (S2).** The file input lacks `capture="environment"`, so Android opens a generic chooser instead of the camera.
- **Full-size photo upload (S2).** The photo is read with `readAsDataURL` and sent as base64 JSON. A 4–8 MB phone photo becomes 5–11 MB on the wire, which is slow on Nepali 3G/4G. Resize client-side to ~1280 px, JPEG quality 0.7 (~150–250 KB).
- **Minus sign on some keyboards (S3).** `type="number"` hides the minus key on some Android keyboards, so negative answers can't be typed. Use `type="text" inputmode="decimal"` with validation.
- **No install or offline story (S4).** No web manifest, `theme-color` or favicon. A small PWA shell would let schools pin it and survive dropped connections.

Mobile layout recommendation: sticky bottom action bar holding the primary button; header reduced to a 48 px bar; photo capture as the first, largest control; Student ID moved to a one-time profile step.

Performance & technical UX

Locally the app is very fast — first paint at 84 ms, every API call under 30 ms — but those numbers come from the mock provider. The UI has no design for the multi-second waits that real OCR and LLM calls will bring.

Metric (local, mock provider)	Value
Page weight	15.7 KB, one HTML file, zero external requests
Time to first byte / first contentful paint	11 ms / 84 ms
API latency (`/diagnose` , `/tutor` , `/practice` , `/assessment`)	5–30 ms
Console errors	None

Findings

- **"Give me a new question" is never new (S2).** The practice call is hard-coded to `index=1`, so a student who repeats the loop gets the same equation every time.
- **No waiting design (S2).** No visible spinner, no progress text, no timeout, no cancel and no retry button. Add staged messages ("Reading your handwriting…" → "Checking each step…") and a 20 s timeout with a retry.

- **Serial, user-gated LLM calls (S3).** The tutor explanation is requested only when the student taps on screen 3. Prefetch it as soon as the diagnosis returns; the student's reading time hides the latency.
- **State is lost on refresh (S3).** Progress through the 7 screens lives in memory only; a refresh or "Start over" returns to defaults. Keep the current screen and inputs in `sessionStorage`.
- **Safe rendering (pass).** Student-supplied steps are written with `textContent`, so typed input cannot inject HTML.

Prioritized issue log

27 issues: 4 critical, 16 major, 5 minor, 2 enhancements. 18 of the 27 are small (under a day each); fixing the 4 critical items first changes what a student or judge takes away from the demo.

#	Severity	Issue	Where	Fix	Effort	Status
1	S1 Critical	Misconception is canned: a sign error, a one-sided step, even "hello" is diagnosed "forgot to divide" with an explanation about 3x = 15	Screens 2–4	Detect the 3 misconceptions deterministically from the parsed steps; never demo on the mock provider	M	Open
2	S1 Critical	"Mastery: YES" shown directly under a wrong answer	Screen 7	Mastery = 3 correct in a row on this skill; show this attempt and overall progress separately	S	Open
3	S1 Critical	Wrong practice answer reveals the solution and ends the loop; no retry	Screens 6–7	Hint → retry (2 tries) → worked solution → new question	M	Open

#	Severity	Issue	Where	Fix	Effort	Status
4	S1 Critical	Diagnosis, result and all UI chrome stay English when Nepali is selected	All screens	Move every string into an en/ne table	M	Open
5	S2 Major	Unreadable input gets a confident diagnosis (72%)	Screen 2	Parse-check each line; show "I couldn't read that" with a retake option	S	Open
6	S2 Major	Steps that don't match the selected problem are diagnosed anyway	Screen 1	Compare line 1 with the problem, or infer the problem from line 1	S	Open
7	S2 Major	Correct work is routed through "What did I get wrong?" and "The misconception"	Screens 2–3	Branch: correct → praise → practice	S	Open
8	S2 Major	Tutor's follow-up question has no answer box; practice then serves a different equation	Screens 4, 6	Answer the follow-up inline, or make it the practice item	S	Open
9	S2 Major	Wrong step not highlighted in the detected steps	Screen 2	Amber marker + one-line note on the faulty line	S	Open
10	S2 Major	Loading text invisible (1.17:1) inside the busy button; no timeout or retry	All async actions	Visible status line, staged messages, 20 s timeout + retry	S	Open

#	Severity	Issue	Where	Fix	Effort	Status
11	S2 Major	Raw errors: "422 — Provide image_base64 or steps_text", "[object Object]"	Screens 1, 6	Validate before submit; friendly inline messages	S	Open
12	S2 Major	Demo steps pre-filled and sent even when a photo is attached	Screen 1	Photo / Type tabs, empty by default; "Try an example" button	S	Open
13	S2 Major	"Give me a new question" always returns the same item (`index=1`)	Screens 5–6	Rotate or randomise the index; track seen items	S	Open
14	S2 Major	6 WCAG 2.1 AA failures: borders, lang, live regions, focus, colour-only stepper, error identification	Global	Apply the Accessibility fixes table	M	Open
15	S2 Major	Red means action, current step and wrong — a "red pen" feel	Global	Crimson in the header only; blue CTA; amber for errors	S	Open
16	S2 Major	No Devanagari font; Nepali styled as a small grey hint	Global	Noto Sans Devanagari; language-aware text hierarchy	S	Open
17	S2 Major	Developer chrome visible: provider tag, confidence %, demo tip, footer	Screens 1–4, footer	Hide behind a debug flag	S	Open
18	S2 Major	Balance visual is static and ignores the student's error	Screen 5	Generate the scale from the equation; tip it with the student's answer	M	Open

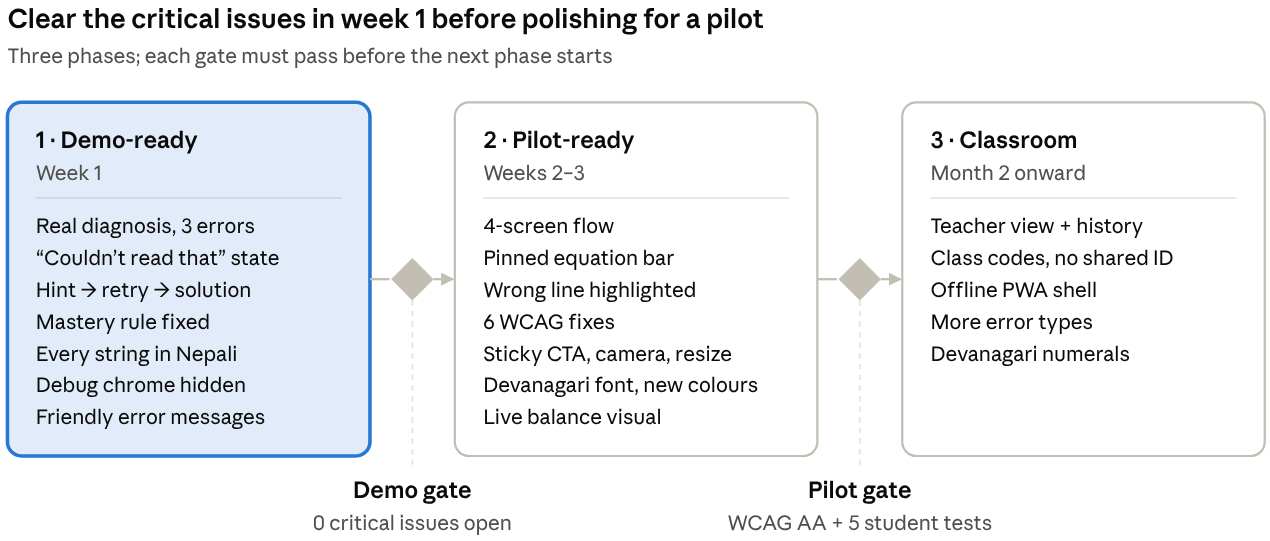
#	Severity	Issue	Where	Fix	Effort	Status
19	S2 Major	Phone: CTA below the fold, 37 px targets, no direct camera, full-size photo upload	Screen 1 (mobile)	Sticky action bar, 48 px targets, `capture="environment"`, client resize to ~200 KB	S	Open
20	S2 Major	Enter / keyboard Go key does nothing	Screens 1, 6	Wrap inputs in a `<form>`	S	Open
21	S3 Minor	7 screens for one loop; 3 screens hold 1–2 lines each	Global	Collapse to 4 screens	M	Open
22	S3 Minor	The original equation disappears after screen 1	Screens 2–7	Pinned equation bar under the stepper	S	Open
23	S3 Minor	Free-text Student ID; everyone shares "demo-student"	Screen 1	Name or class code once, stored on the device	M	Open
24	S3 Minor	Start over reloads the page and resets language and problem	Screen 7	Keep choices; return to screen 1	S	Open
25	S3 Minor	Polish: list numbers outside the box, spell-check squiggles under "3x", `type=number` hides minus on some keyboards	Screens 1, 2, 6	`list-style-position: inside`; `spellcheck="false"`; `inputmode="decimal"`	S	Open
26	S4 Enhancement	No history or teacher view, though the API returns recent activity	New screen	Simple timeline for student and teacher	L	Open

#	Severity	Issue	Where	Fix	Effort	Status
27	S4 Enhancement	No manifest, favicon or offline shell; no Devanagari-numeral option	Global	Small PWA shell; numeral toggle	M	Open

Effort: S = under a day, M = 1–3 days, L = a week or more.

Recommendations & roadmap

Spend week 1 only on the 4 critical issues plus the cheap trust fixes; nothing else matters if the diagnosis is wrong or the result contradicts itself.



roadmap · 3 phases, 2 gates

Phase 2 ends with think-aloud tests: 5–6 Grade 7–10 students in Nepali, each solving 2 problems on their own phone, with a teacher watching.

Success metrics

Metric	Today	Target
Wrong inputs given the right misconception label (audit set of 4)	1 of 4	⩾ 90% on a 50-item labelled set
Taps from submit to practice	5	3

Metric	Today	Target
UI strings shown in Nepali when Nepali is selected	Tutor paragraph + 2 hints	100%
WCAG 2.1 AA failures	6	0
Second-attempt success after a hint	Not measurable (no retry)	Tracked; ≥ 60%
Photo upload size on a phone	Full photo, 5–11 MB as base64	≤ 250 KB
Students who finish the loop in testing	Not yet tested	≥ 5 of 6